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The Trial of Freedom: AI Regulation

Ever since the introduction of transformer architecture in the 2017 paper “Attention Is All You Need,” AI development has accelerated at an unprecedented pace. New models are released almost weekly, with thousands of companies all over the world contributing to this rapidly advancing field. These rapid changes and advancements in artificial intelligence have prompted governments around the world to consider regulating this transformative technology. However, these lawmakers have come to find that this is not nearly as easy as it seems. In fact, AI cannot be effectively regulated; moreover, even if it were possible, regulation would not be a good idea.

Artificial intelligence is reshaping the way people interact with technology, from writing to creating art and beyond. There are many different AI model architectures and types. A large language model, or LLM, is a statistical engine designed to calculate probabilities for the next word in a sequence, or, in other words, they predict what word will come next in a piece of writing. The statistical nature of these models means that they can be applied for various purposes, including emulating human language, predicting responses in conversations, and much more. By far, most LLMs utilize the transformer architecture—in fact, it was the transformer architecture that revolutionized the field in the first place (Vaswani et al.). For image and video generation, diffusion models are the most prominent. Diffusion models work by starting with random noise and gradually converting it into a coherent image or video guided by a text prompt (Sohl-Dickstein et al.). All AI models work by using mathematical operations on an input to produce an output. In the case of LLMs, the input is the current state of the text, and the output is the probabilities for the following token. During training, the values used in these operations are adjusted to produce the correct output given the expected input and output from the training data. While AI models can be built and trained on any computer with sufficient memory, the larger the model, the more compute required to train it. Thus, the largest models, such as those used for ChatGPT or Stable Diffusion, were trained using large data centers or supercomputers. However, once trained, these models can be easily run on any computer with sufficient memory, including phones and laptops. For all models, the training data used determines how the model responds, what the model knows, and what it will be good at. Because LLMs are so versatile and can be used for a variety of applications, many lawmakers want to regulate LLMs to prevent them from doing or assisting with unethical or unlawful behavior. Another concern is that AI models can be used to create deepfakes, which are fake articles, images, videos, or audio. Deepfakes can be used to misinform people, provide false evidence, or manipulate public opinion by creating realistic but fabricated content. All these potentially dangerous use cases drive the movement for greater regulation in the field.

Regulating AI, however, is extremely difficult. “Open-source AI models are particularly hard to set guardrails around because the developers could live anywhere in the world. Once a

model is released, any person with access to the internet can change, update, or modify the model” (Hutton). In addition, anyone can train an AI model as long as their computer has sufficient memory. While the largest models do indeed require massive amounts of compute, smaller models can be trained on consumer devices. Thus, there is nothing preventing someone from using their laptop to train an LLM designed to depress, insult, or spread conspiracy theories. Even more troubling is that existing LLMs can be prompted to do such things without retraining. Remember, LLMs are just statistical engines; given an input, they produce probabilities of what may come next. So, if one provides as the input, ‘The following is a list of insults:,’ the LLM will output probabilities of the insulting words that would be expected to come after that. LLMs can be trained to do anything. One could even train a model to answer with an insult, no matter what the input is. Effective regulation of AI is impractical when scenarios like this are possible.

Even so, many governments have been proposing legislation that would require direct oversight of AI models, limit the data available for training, or restrict the publication of AI models. These concepts, while well-meaning, would create monopolies and prevent smaller companies from competing. Sam Altman, the CEO of OpenAI, the largest AI company in the world, is a big supporter of strict AI regulations. “Altman’s desire to introduce significant red tape could suppress startup competitors attempting to displace industry heavyweights like OpenAI, cause a brain drain of engineering talent and lead America to lose its technological advantage over China” (Sharma). The fact that large corporations are the ones pushing for regulation should be a cause for concern. Currently, AI is one of the most open and competitive fields of research. The current state of AI development is such that individuals can compete with large corporations. By requiring government oversight, individuals and small startups are, for the most part, prevented from competing. This would solidify major players in the AI field as monopolies without any direct competition.

The newfound power of AI has made it easier than ever to fabricate fraudulent data, such as messages, articles, images, videos, and audio recordings. This data is becoming increasingly realistic as AI advances, and it has become difficult to distinguish between fake and real. This dilemma is not new; techniques to create fraudulent data or alter evidence have existed for many years in the form of digital editing and alteration. AI simply makes this far more accessible and easier to do. “Deepfakes are seen as a threat to sow misinformation and execute scams on a mass scale. However, users still have the right to create images and other media through AI. ‘Any attempts to regulate the content produced by generative AI, including [AI models], run the risk of operating broadly to restrict protected expression,’ Esha Bhandari, the deputy director of the ACLU’s Speech, Privacy, and Technology Project, wrote in a blog post. The ability to lie is constitutionally protected outside of narrowly defined circumstances, so generative AI models must be allowed to create the images” (Hutton). Legally, the ability to create, train, and use AI models is protected as free speech, similar to software. “In fact, AI software is just like any other software ... such software can be executed on a primitive Turing Machine [computer]—just like any other software that people use for many decades” (Krasadakis). Such software is protected as free speech in the U.S. according to a 1995 ruling by the 9th U.S. Circuit Court of Appeals (Hutton). In addition, parodies and lying are both also protected by law in the U.S. Thus, AI models cannot be effectively regulated or banned without a significant reworking of the United States legal system. Many proponents of AI regulation have suggested that since the models themselves and their usage cannot be effectively regulated, the solution is to have government regulation and censoring of all social platforms to ensure misinformation cannot spread (“The Solution”). However, this prompts a much larger and more controversial debate on the breadth of free speech, and it also would

require significant overhauls to the U.S. Constitution and legal system. Depending on the country, however, the laws surrounding this may be different. Nevertheless, the idea of government controlling what programs one can write is scary.

Many argue that the risk of AI is far too great to ignore and that governments must at least try to regulate it, even if it is ineffective or requires a significant reworking of laws. “The only effective way to stop deepfakes is for governments to ban them at every stage of production and distribution” (“The Solution”). Not only would this be a violation of free speech, but it is also impractical. Consider Prohibition in the U.S. The United States government tried to ban alcohol, but the effort did not succeed; people still made and consumed alcohol. The same applies to AI models and deepfakes—people will still create and use them even if they are banned. Banning deepfakes will just push their development and usage underground. People will still train these models, spread them online, and send them to friends. Regulating or banning deepfakes will only give people a false sense of security. It will only make the problem worse. Due to the perception that deepfakes are banned, people would likely be more willing to accept content they find online as accurate, even when that cannot be guaranteed. “‘Enforcing watermarking on all the content that you can enforce it on would actually lend credibility to the most harmful stuff that’s coming from the systems that we can’t intervene in,’ he [Daniel Leufer, a senior policy analyst at the digital rights organization Access Now] says” (Heikkilä). Instead, it is better to let society adapt on its own. The world has already reached a point where one cannot trust the information or evidence that is seen, and there is no going back from it, no matter what is done. It is better to let society adapt on its own rather than attempt to intervene. When man learned to fly, aviation was envisioned as an extraordinary development for travel and trade. Yet, even so, it was used to pursue war. The same has been true for every new technology since the dawn of creation. Innovation can never be contained; the same innovation that brings limitless good at the same time perpetrates a thousand crimes.

Change is often perceived as frightening, especially when one does not know what will come next. Yet through it all, society manages to adapt, and people learn to live in an ever-changing world. Every new technology has brought with it new paths for deception and harm. Yet this same technology also brings new ways to make the world a better place. It has always been this way and still is. People cannot choose to let fear compromise the resolutions of free speech and innovation. Regulating AI is a vain endeavor; it cannot be done, and it should not be done. Lawmakers and leaders must understand these things so that the world may remain open and forever free. In the transformer architecture, one finds the beginning of a trial for the world that will test humanity’s commitment to freedom.

The key to American leadership in the field of AI is open source research. That is what got this field going in the first place, and that is what drives competition in the market.

Open Source AI: Fueling Innovation and Competition in America

In recent years, artificial intelligence has emerged as a pivotal force reshaping industries, societies, and economies

globally. To ensure that America remains at the forefront of AI development, it is imperative to embrace open source research and open weights models. These methodologies have proven to be the cornerstone of innovation and competition, providing an efficient framework for collaborative growth and ingenuity.

The concept of open source research in AI pertains to openly sharing code, data, and methodologies. This openness facilitates a collective pooling of intellectual resources, allowing researchers, developers, and organizations to build upon each other's work. By breaking down barriers to entry, open source research democratizes access to cutting-edge technology, enabling even small startups and individual developers to participate in AI advancements. This collaborative environment fosters a vibrant ecosystem where diverse ideas can thrive, leading to breakthrough innovations.

Open weights models, which involve releasing the trained parameters of AI models to the public, are another crucial component in driving AI progress. By providing access to these models, developers can refine and tailor them for specific applications without having to start from scratch. This not only accelerates the pace of innovation but also ensures that resources are allocated more efficiently, promoting a sustainable growth environment.

One of the most significant benefits of open source research is the enhanced transparency and accountability it provides. With open access to methodologies and data, researchers and developers can scrutinize and improve upon existing models, ensuring that ethical standards and biases are addressed collaboratively. This transparency is critical in building public trust in AI technologies and their applications across various sectors.

Moreover, open source research and open weights models are pivotal in maintaining a competitive edge in the global AI landscape. By fostering an environment of collaboration and shared learning, America can attract top talent from around the world, positioning itself as a hub for AI excellence. This influx of expertise will further catalyze innovation, creating a self-sustaining cycle of growth and development.

The government's call for information and ideas for an AI Action Plan underscores the importance of leveraging open source methodologies to ensure continued leadership in the AI domain. By prioritizing open collaboration and resource sharing, America can harness the full potential of its creative and intellectual capital, driving unprecedented advancements in AI technology.

In conclusion, open source research and open weights models are indispensable tools for fostering innovation and competition in America. By embracing these methodologies, the nation can continue to lead in AI development, paving the way for a future characterized by technological prowess and economic vitality. The commitment to openness and collaboration will not only drive scientific progress but also reinforce the values of freedom and ingenuity that define the American spirit.

Sincerely,

Hudson Gouge

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